Program : **Diploma in Artificial Intelligence & Engineering**	
Course Code : **4381**	Course Title: **Fundamentals of Machine Learning**
Semester : 4	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- Students will be able to get basic knowledge on the techniques to build an intellectual machine for making decisions on behalf of humans.
- This course also covers the techniques on how to make learning by a model, how it can be evaluated, what are all different algorithms to construct a learning mode.

Course Prerequisites:

Topic	Course code	Course name	Semester
Mathematics	1002,2002		1,2
Introduction to IT systems Lab	1008		1
Problem Solving and Programming			2
Introduction to Artificial Intelligence			3

Course Outcomes

On completion of the course, the students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand Machine Learning concepts	15	Understanding
CO2	Familiarize Supervised Machine Learning Concepts - Classification and Regression	14	Understanding

CO3	Familiarize clustering, dimensionality reduction and reinforcement learning	14	Understanding
CO4	Demonstrate evaluation measures of machine learning models	15	Understanding
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Machine Learning concepts**	15	Understanding
M1.01	Explain Machine Learning	3	Understanding
M1.02	Illustrate some examples in Machine Learning Applications, Life cycle of ML	3	Understanding
M1.03	Explain datasets -How to get datasets – Types of Data – Types of Datasets, feature sets, dataset division	3	Understanding
M1.04	Summarize Data Preprocessing in Machine learning	4	Understanding
M1.05	Explain Machine Learning Techniques	2	Understanding

Contents:

Machine Learning - Definition - History- Features- Classification of ML ,

Machine Learning Applications- Life cycle of ML - Process involved in Machine Learning Techniques.

Datasets -How to get datasets – Types of Data – Types of Datasets, feature sets, dataset division,

Data Preprocessing in Machine learning – Steps for data preprocessing

Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning

CO2	**Familiarize Supervised Machine Learning Concepts - Classification and Regression**	14	
M2.01	Explain Regression analysis	2	Understanding
M2.02	Explain Linear Regression	2	Understanding
M2.03	Explain the concept of Backward Elimination, Polynomial Regression	3	Understanding
M2.04	Explain the concept of Classification Algorithm in ML	2	Understanding
M2.05	Illustrate the Types of ML Classification Algorithms	5	Understanding
	Series Test – I	1	

Contents:

Supervised Learning- Classification and Regression

Regression analysis- Terminologies-Types of regression,

Linear Regression -Types of Linear Regression- Simple Linear Regression - Multiple Linear regression -Applications

Backward Elimination – Basic concept of Backward Elimination - Steps of Backward Elimination , Polynomial Regression - Steps for Polynomial Regression

Classification Algorithm in Machine Learning - Learning a Class from Examples**,**

Types of ML Classification Algorithms - Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest

CO3	**Familiarize clustering, Dimensionality reduction and Reinforcement learning**	14	Understanding
M3.01	Understand concepts and types of Clustering	4	Understanding
M3.02	Explain Dimensionality reduction	4	Understanding
M3.03	Discuss features and types of reinforcement learning	4	Understanding
M3.04	Discuss Q learning	2	Understanding
Contents: Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering and Divisive clustering Partitional Clustering - K-means clustering-Mean Shift clustering. Dimensionality reduction – Principal Component Analysis, factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Reinforcement learning: Key features- types - Markov Decision Process-Q learning			
CO4	**Demonstrate evaluation measures of machine learning models**	**15**	Understanding
M4.01	Discuss regression- Mean Absolute Error, Mean Squared Error	3	Understanding
M4.02	Discuss regression- Root Mean Squared Error, R-squared.	3	Understanding
M4.03	Explain evaluation measure classification	3	Understanding
M4.04	Discuss Clustering: Sum of Squared Error, Silhouette Coefficient , Dunn's Index Ensemble methods	3	Understanding
M4.05	Discuss Clustering: Bootstrapping, Cross Validation	3	Understanding
	Series Test – II	1	

Contents: **Evaluation Measures:** Regression: Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, R-squared. Classification: Accuracy, confusion matrix, precision, recall, F-Score, ROC-Curve. Clustering: Sum of Squared Error, Silhouette Coefficient, Dunn's Index Ensemble methods, Bootstrapping, Cross Validation.

Text / Reference:

T/R	Book Title / Author
T1	Kevin P. Murphy. Machine Learning: A Probabilistic Perspective, MIT Press 2012.
T2	Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010.
R1	Mitchell M., T., Machine Learning, McGraw Hill (1997) 1stEdition.
R2	Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., Machine Learning, Neural and Statistical Classification. Overseas Press (1994)

Online Resources:

Sl. No.	Website Link
1	https://www.javatpoint.com/machine-learning-models
2	https://www.javatpoint.com/machine-learning
3	Introduction to Machine Learning - Course (nptel.ac.in)